

# A Day Before the Flood: Terrain-Aware Neural Forecasting of River High-Flow Events in Sri Lanka

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**Abstract**—This paper evaluates MMF-Net, a terrain-conditioned temporal transformer for next-day high-flow prediction at 51 Sri Lankan river-basin locations. It encodes 14-day sequences of 33 dynamic variables with periodic numerical embeddings, temporal and cross-feature attention, and nine static attributes supplied through feature-wise linear modulation. The target is exceedance of a training-period Global Flood Awareness System (GloFAS) discharge percentile, not observed inundation. In recorded temporal experiments, periodic embeddings increase test average precision (AP) from 0.6083 to 0.7605; a terrain-conditioned five-seed binary-cross-entropy ensemble reaches AP 0.8355, Brier score 0.0080 and expected calibration error 0.0016. Results support further controlled study of numerical representation and static conditioning. Interpretation is limited by single-seed ablations, incomplete radar controls and proxy supervision.

**Index Terms**—MMF-Net, next-day high-flow classification, periodic numerical embeddings, terrain-conditioned transformer, GloFAS discharge reanalysis, Sri Lankan river basins

## I. INTRODUCTION

Predicting high river flow requires information about both recent weather and the state of the catchment. Similar rainfall totals can produce different responses depending on antecedent wetness, existing discharge and local conditions. A neural forecasting model must therefore represent interactions among variables as well as their evolution through time. In Sri Lanka, a dataset spanning contrasting climatic zones provides an opportunity to study these representations across multiple river-basin locations.

Recurrent neural networks have demonstrated the value of learning rainfall–runoff relationships from hydrological sequences [1]. More recent large-scale work has extended neural flood forecasting to ungauged watersheds [2]. These results motivate regional studies, but the distinction between the prediction target and the intended application remains essential. A network trained to reproduce a discharge reanalysis is evaluated against that reanalysis. Its performance cannot be interpreted directly as accuracy against measured water levels, inundation extent or damage.

This paper investigates MMF-Net, a deep neural network (DNN) combining a temporal transformer, periodic numerical embeddings, cross-feature attention and static conditioning. It documents the implemented architecture and recorded ablations while separating them from non-matched historical baselines.

Three research questions (RQs) guide the analysis: (RQ1) how do numerical embeddings, cross-feature attention and terrain conditioning affect the recorded neural prediction scores; (RQ2) how do the training objective and ensemble calibration affect ranking and probability quality; and (RQ3) what evidence is available for radar fusion and practical warning use?

## II. CONTRIBUTIONS

This study makes three bounded contributions. First, it documents a terrain-conditioned neural architecture that combines periodic numerical embeddings, temporal and cross-feature attention, static feature-wise linear modulation (FiLM), and an explicitly optional radar branch. Second, it reports the recorded chronological ablations on one consistent main DNN snapshot, including the effects of numerical representation, feature interaction, static conditioning, training objective, ensembling, and temperature scaling. Third, it distinguishes reanalysis-based high-flow ranking from observed-flood or operational-warning performance, and records the evidence that remains needed: prediction arrays, matched baselines, uncertainty estimates, issue-time inputs, and independent gauge or inundation validation.

## III. RELATED WORK

### A. Neural hydrological forecasting

Following neural rainfall–runoff modelling [1], large-scale studies have demonstrated prediction of extreme river flows in ungauged watersheds [2]. Here the task is daily classification at fixed Sri Lankan locations using reanalysis supervision, which differs from operational global forecasting in its scale, target and evaluation.

Local research also motivates neural temporal models. Saubhagya *et al.* combine a bidirectional LSTM with time-series regression for water-level forecasting in the Ratnapura area of the Kalu basin [3]. Differences in target, horizon and data prevent direct comparison with the present multi-location exceedance task.

### B. Representation, conditioning and spatial structure

Transformers use self-attention to represent relationships among sequence elements [4]. For continuous inputs, Gorishniy *et al.* show that feature embeddings can improve deep

models across several backbones [5]. We apply their periodic embedding approach before temporal encoding. Feature-wise linear modulation (FiLM) conditions a learned representation on auxiliary attributes [6]; here these describe terrain, climatic zone and network position.

River graphs offer another way to incorporate hydrological structure, although their benefit must be established against a suitable per-location control. Kirschstein and Sun found that their neural forecasting models did not benefit from the supplied river topology [7]. This motivates a controlled question rather than a presumption that adding edges improves forecasting. MMF-Net is graph-free: weights are shared across locations, but predictions do not exchange messages between them.

### C. Probability quality and evaluation

Focal loss was introduced to address the dominance of easy examples in dense object detection [8]; its suitability for hydrological classification is an empirical question. Deep ensembles and temperature scaling provide established approaches to probability estimation and post-hoc calibration [9], [10]. Under rare positive events, precision-recall analysis complements threshold-based verification [11]. Temporally structured observations also require an evaluation design that respects their dependence [12].

## IV. STUDY DATA AND PREDICTION TASK

### A. Spatial coverage and input provenance

The inventory has 51 locations in 16 basin or area groups: 31 wet-zone, 13 dry-zone and seven intermediate-zone nodes. Fig. 1 shows their coordinates and 35 curated directed flow links. These are modelling locations, not 51 measured-discharge gauges; lines indicate connectivity rather than surveyed channels.

NASA POWER supplies base daily meteorological and soil-wetness variables [16]. The feature builder preferentially uses cached Open-Meteo Archive precipitation where available, falling back to POWER precipitation. Discharge is obtained through the Open-Meteo Flood API from the GloFAS product family [17], [18]. Product versions and precipitation-source proportions should be retained in a data manifest before replication.

The 33 dynamic inputs comprise rainfall and antecedent-rainfall variables, temperature, humidity, wind and radiation variables, soil-wetness variables, and discharge variables. Discharge features include recent changes, rolling means, seasonal anomalies and the empirical percentile. The nine static inputs comprise elevation, log mean-training-discharge as a drainage proxy, three climatic-zone indicators and four network-position indicators. The drainage proxy is not a measured catchment area.

Selected heavy-tailed dynamic variables undergo a signed  $\log(1 + |x|)$  transformation followed by standardisation. Dynamic statistics are fitted on the temporal training subset, while discharge percentile references and climatologies are fitted through 2017. Static continuous attributes are standardised

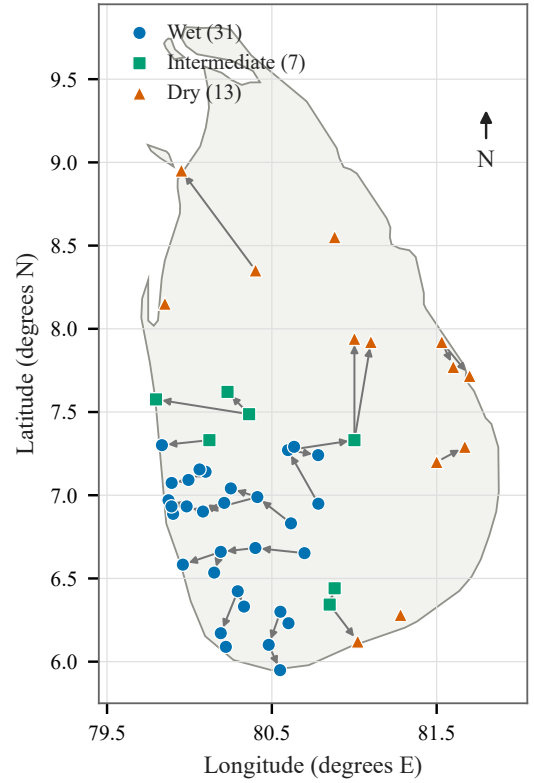


Fig. 1. Study locations and curated upstream-to-downstream links. Colours and marker shapes identify the inventory’s climatic-zone categories. Country outline: Natural Earth. Hydrological basin boundaries are not shown.

across the node inventory. Missing normalised dynamic values are replaced with zero, corresponding to the training mean; the main tabular stream has no separate missingness mask.

### B. Targets and episode definition

Let  $Q_{n,t}$  denote reanalysis discharge at location  $n$  on day  $t$ . The high-flow state and primary next-day target are defined in (2):

$$\tau_n = \text{quantile}_{0.98}\{Q_{n,t} : t \leq 2017-12-31\}, \quad (1)$$

$$s_{n,t} = \mathbf{1}[Q_{n,t} \geq \tau_n], \quad y_{n,t}^{(1)} = s_{n,t+1}. \quad (2)$$

This percentile of daily discharge is a relative high-flow threshold, not a calibrated return-period estimate or an official warning-stage threshold. It does not establish whether water leaves the channel or affects settlements.

Auxiliary classification targets indicate any exceedance in the next two and three days,  $y_{n,t}^{(h)} = \max_{k=1,\dots,h} s_{n,t+k}$ , and next-day onset,  $y_{n,t}^{(o)} = (1 - s_{n,t})s_{n,t+1}$ . Two regression heads predict the next-day discharge percentile and the maximum discharge z-score over the next three days. All reported main-model scores concern  $y^{(1)}$ , not the onset head.

Episodes group positive state days at each node when successive dates are at most two days apart. They are node-level proxy episodes, not independent national disasters.

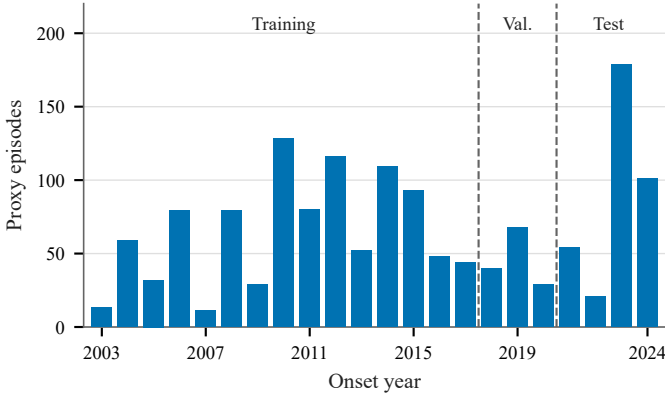


Fig. 2. Annual counts of node-level proxy episodes with onset in 2003–2024, computed from the supplied catalogue (1,464 entries). Divisions indicate the temporal training, validation, and test periods. These are catalogue counts, not the episodes eligible for a particular model evaluation.

### C. Coverage and data-version scope

Inputs end on 31 December 2024. The supplied catalogue has 1,464 proxy episodes starting in 2003–2024. Fig. 2 records the retained catalogue; its annual counts should not be interpreted as a trend in observed disasters.

Exact post-filter sample counts and saved predictions are absent; historical totals extending into 2025 are not used to describe the truncated DNN experiment.

## V. DEEP NEURAL NETWORK METHOD

### A. Numerical and temporal representation

For a prediction issued after day  $t$ , the input is  $\mathbf{X}_{n,t} \in \mathbb{R}^{14 \times 33}$ , ending on that day. Nodes are folded into the batch dimension, so the same network processes every node’s history independently. For scalar feature  $j$ , the periodic–linear–ReLU (PLR) representation is defined by (3):

$$\mathbf{e}_j(x) = \text{ReLU}(\mathbf{W}_j[\sin(2\pi\mathbf{c}_jx) \parallel \cos(2\pi\mathbf{c}_jx)] + \mathbf{b}_j), \quad (3)$$

where  $\mathbf{c}_j \in \mathbb{R}^8$  contains trainable frequencies and  $\mathbf{e}_j \in \mathbb{R}^8$ . This supplies a flexible nonlinear representation before features are mixed; it does not imply that an ordinary deep network cannot learn nonlinear boundaries.

Each day’s feature embeddings are concatenated and projected to a 128-dimensional token. Learned position embeddings and a classification token are passed through three pre-layer-normalised transformer blocks, each with four attention heads, a feed-forward width of 256 and dropout of 0.2. The final classification-token state is the temporal representation  $\mathbf{h}^T$ .

### B. Cross-feature attention and static conditioning

The feature branch uses the same embeddings, averaged over the 14-day window. Each feature is projected to a 128-dimensional token with a learned feature identity. A one-layer transformer pools these tokens to produce  $\mathbf{h}^F$ . The branches merge additively as shown in (4):

$$\mathbf{h} = \text{LN}(\mathbf{h}^T + \mathbf{h}^F). \quad (4)$$

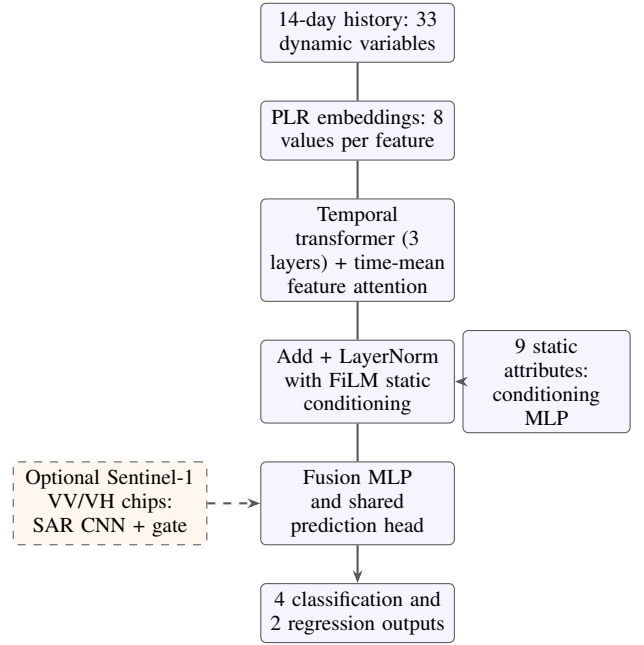


Fig. 3. MMF-Net data flow. Temporal and cross-feature attention are parallel branches sharing numerical embeddings. Static attributes condition their merged state. The principal configuration uses the solid path; optional radar fusion is applied before the fusion MLP. No graph message passing is used.

This branch explicitly attends across feature tokens. Averaging first reduces its computational cost but removes observation order within each feature token. Here and below, LN denotes layer normalisation.

A multilayer perceptron (MLP) maps the static vector  $\mathbf{s}_n \in \mathbb{R}^9$  to  $\gamma_n$  and  $\beta_n$ , giving the conditioned representation in (5):

$$\tilde{\mathbf{h}} = \text{LN}[\mathbf{h} \odot (1 + \gamma_n) + \beta_n]. \quad (5)$$

The final conditioning layer is zero-initialised, so the initial affine modulation is neutral. Layer normalisation still applies; the full FiLM block is not an identity map. A fusion MLP of widths 192 and 128 and a shared head of widths 256 and 128 produce classification logits and regression outputs. Fig. 3 shows the implemented data flow.

### C. Optional synthetic-aperture radar branch

The convolutional neural network (CNN) variant encodes two-channel Sentinel-1 vertical-transmit/vertical-receive (VV) and vertical-transmit/horizontal-receive (VH) chips and adds their projection through a learned gate. For radar embedding  $\mathbf{v}$ , presence indicator  $m$  and image age  $a$ , the fusion is specified in (7):

$$\mathbf{g} = \sigma\{\text{MLP}[\tilde{\mathbf{h}} \parallel \mathbf{v} \parallel m \parallel (a/30)]\}, \quad (6)$$

$$\mathbf{h}^S = \text{LN}(\tilde{\mathbf{h}} + m \mathbf{g} \odot \mathbf{W}_v \mathbf{v}). \quad (7)$$

The gate’s final bias is initialised to  $-3$ , giving a small, nonzero opening; absent frames contribute zero through  $m$ . A scalar variant instead appends four image-derived summaries to the dynamic stream. Experiment notes report imagery at nine of 51 nodes. Pretraining support exists in the code, but a

TABLE I  
MAIN TEMPORAL DNN CONFIGURATION.

| Setting                    | Value                         |
|----------------------------|-------------------------------|
| Training / validation      | 2003–2017 / 2018–2020         |
| Test input dates           | 2021–2024                     |
| Input dimensions           | 14 days; 33 dynamic, 9 static |
| Temporal / feature blocks  | 3 / 1; width 128, 4 heads     |
| Batch size                 | 32 daily snapshots; CNN: 8    |
| Optimiser / maximum epochs | AdamW / 60                    |
| Seeds                      | N0–N4: 1; N5 and N6: 5        |
| Main target                | Next-day discharge exceedance |
| Checkpoint / threshold     | Validation episode AP / F1    |

completed pretrained checkpoint is not verified in the recorded evidence. The CNN result is therefore not presented as a validated pretrained-radar experiment.

#### D. Multi-task optimisation and calibrated inference

The reference objective sums masked binary cross-entropy (BCE) over four heads, with weights (1.0, 0.3, 0.3, 0.5), and adds the mean Huber loss over the two regressions with weight 0.2 [13]. The alternative `focal_conf` objective uses focal loss with  $\alpha = 0.75$ ,  $\gamma = 2$ , and a label-confidence weight between 0.5 and 1 that decreases near the current-day discharge threshold. The comparison changes class weighting, focusing and confidence weighting jointly; it is not an isolated test of the focal exponent.

The implementation uses PyTorch [14]. AdamW [15] uses a learning rate of  $10^{-3}$ , weight decay  $10^{-4}$ , five warmup epochs and a cosine schedule. Training lasts at most 60 epochs, with gradient norm clipping at 1.0 and early-stopping patience of ten. Checkpoints are selected by validation episode-level average precision, falling back to node-day average precision if the episode score is undefined.

For five-seed ensembles, the implementation averages predicted probabilities, not logits. If  $p_k$  is the probability from seed  $k$ , calibration uses (8):

$$\bar{p} = \frac{1}{5} \sum_{k=1}^5 p_k, \quad p_{\text{cal}} = \sigma \left[ \frac{\logit(\bar{p})}{T} \right]. \quad (8)$$

The temperature  $T$  is fitted on validation data. The decision threshold is selected by validation F1 and frozen for test evaluation. The reported operating points do not use a common false-alarm budget.

## VI. EXPERIMENTAL DESIGN

### A. Main DNN ablations and historical references

Table I gives the recorded temporal protocol and operational settings. These values describe the stored run snapshot, not a new rerun.

The temporal split is 2003–2017 for training, 2018–2020 for validation and 2021–2024 for testing. Inputs contain 14 days of 33 dynamic and nine static variables. N0–N3 successively introduce PLR embeddings, feature attention and FiLM; N4 changes to `focal_conf`; N5 adds a five-seed ensemble and temperature scaling; and N5\_bce applies the same ensemble

TABLE II  
MAIN DNN RESULTS ON THE TEMPORAL TEST SPLIT. AP: AVERAGE PRECISION; ECE: EXPECTED CALIBRATION ERROR; FAR: FALSE-ALARM RATIO; PARAMS: PARAMETER COUNT.

| ID     | Configuration       | AP           | ECE   | FAR  | Params |
|--------|---------------------|--------------|-------|------|--------|
| N0     | Linear embeddings   | .6083        | .0024 | .412 | .551   |
| N1     | + PLR               | .7605        | .0053 | .325 | .555   |
| N2     | + feature attention | .7738        | .0056 | .313 | .693   |
| N3     | + static FiLM       | .8269        | .0032 | .110 | .711   |
| N4     | + focal/confidence  | .7592        | .0525 | .138 | .711   |
| N5     | N4, ensemble + TS   | .7846        | .0043 | .145 | .711   |
| N5_bce | N3, ensemble + TS   | <b>.8355</b> | .0016 | .120 | .711   |
| N6-sc  | SAR scalars, focal  | .7284        | .0055 | .344 | .719   |
| N6-gat | SAR CNN, focal      | .8310        | .0066 | .107 | 11.967 |

TS: temperature scaling. N0–N4: one seed; others: five. Params: millions per member. “Focal”: combined focal/confidence objective. Bold marks the largest recorded AP, not significance. Source: model-2 snapshot, 10–11 August 2026.

to N3 with BCE. N6 radar variants use `focal_conf`, so they are not matched BCE controls. Historical tree and graph runs use a different panel and are not ranked against MMF-Net.

### B. Metrics and their interpretation

The stored metric named precision–recall area under the curve is computed as stepwise average precision (AP),  $\sum_i P_i \Delta R_i$ , rather than a trapezoidal area; we use AP throughout. The implementation sorts individual samples without grouping equal scores, so tied-score references require a standard tie-aware recomputation before direct comparison.

Node-day quantities are probability of detection,  $\text{POD} = TP/(TP + FN)$ ; false-alarm ratio,  $\text{FAR} = FP/(TP + FP)$ ; and critical success index,  $\text{CSI} = TP/(TP + FP + FN)$ . FAR is the fraction of positive predictions that are false alarms, not  $FP/(FP + TN)$ . Brier score averages  $(p - y)^2$ . Expected calibration error (ECE) uses 15 equal-width probability bins, weighted by sample counts. Under class imbalance, a small aggregate ECE can coexist with poor calibration in sparsely populated high-risk bins.

Episode detection credits any alarm at a node during the seven days preceding its episode onset. This is a retrospective association statistic. Because the model predicts the next day, an alarm six days before onset is not evidence of a calibrated six-day forecast. An onset-specific assessment should align the evaluation window with the operational lead time.

## VII. RESULTS

The following analyses address RQ1 through the N0–N3 representation and conditioning ablations, RQ2 through the objective and ensemble comparisons, and RQ3 through the radar and practical-use evidence.

### A. Neural representation and static conditioning

Table II reports the recorded MMF-Net results. PLR embeddings increase AP from 0.6083 to 0.7605, feature attention to 0.7738 and static FiLM to 0.8269. The FiLM contrast combines elevation, drainage proxy, zone and position indicators.

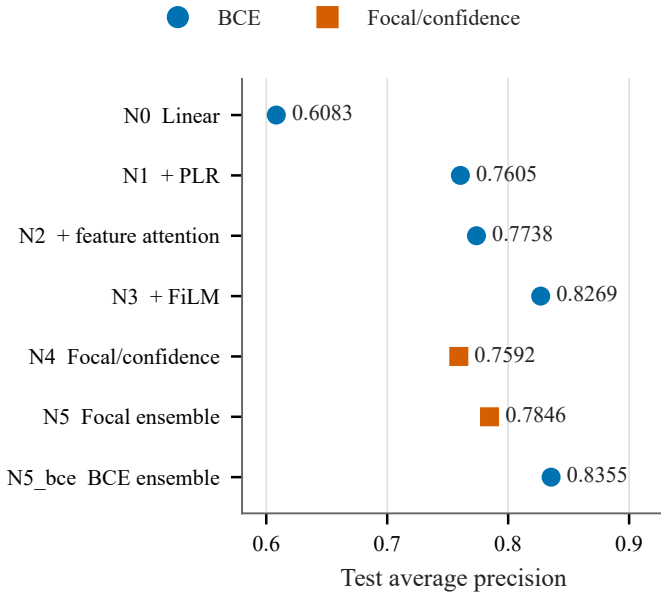


Fig. 4. Recorded DNN average precision. Points show individual configurations. Marker shape identifies the training objective. No confidence intervals are available; the plot does not establish statistical significance.

Fig. 4 shows the same AP values. N0–N4 are single-seed runs, so their differences are descriptive rather than significance-tested effects.

#### B. Objective, ensemble and calibration

The focal/confidence objective changes N3’s AP/ECE from 0.8269/0.0032 to 0.7592/0.0525. N5\_bce records 0.8355/0.0016 versus N5’s 0.7846/0.0043, favouring BCE in these settings. Its Brier score is 0.0080 and CSI is 0.561. Ensembling and temperature scaling change together, so their separate effects are not identified.

#### C. Radar fusion and model size

The scalar radar variant records AP 0.7284. The gated CNN records 0.8310, compared with 0.7846 for the image-free focal/confidence ensemble and 0.8355 for the image-free BCE ensemble. This is mixed evidence: the CNN result is higher under the focal/confidence family, yet does not exceed the strongest recorded image-free result. Differing objectives, batch sizes, uncertain pretraining status and limited coverage prevent attribution to radar alone.

Fig. 5 places this comparison in a resource context. N5\_bce has 0.711 million reported parameters per member; N6-gat has 11.967 million. The five-member image-free ensemble therefore requires five forward passes, while device-level latency and energy remain unmeasured.

#### D. Detection at recorded operating points

Fig. 6 shows episode detection and node-day FAR for selected DNN configurations. N2 detects 0.423 of episodes at FAR 0.313; N3 records 0.197 at FAR 0.110; and N5\_bce records 0.220 at FAR 0.120. Each point uses its own

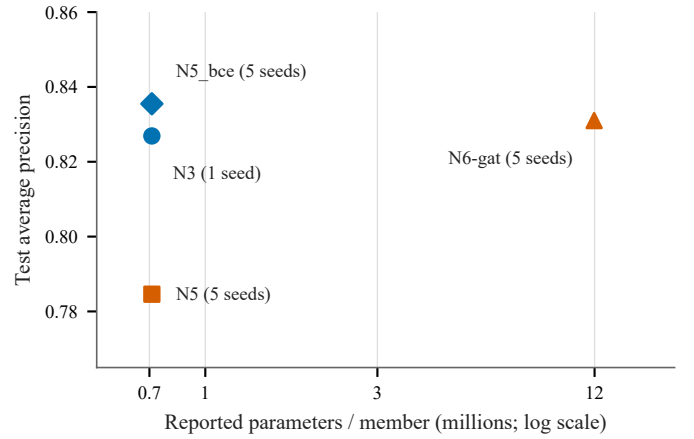


Fig. 5. Reported parameter count per member and AP for selected DNNs. The horizontal axis is logarithmic; parameter count does not measure latency or the value of radar information.

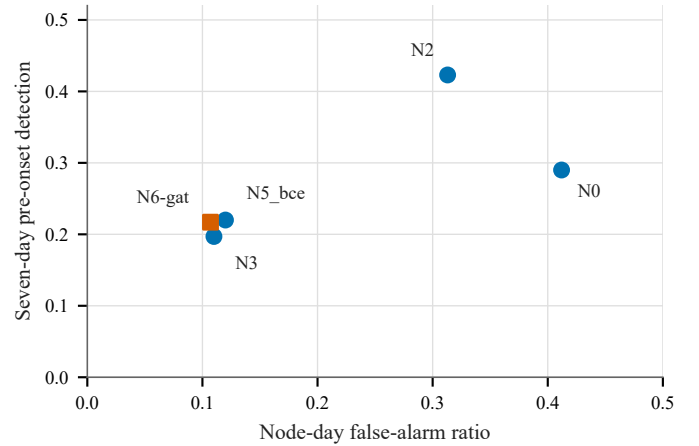


Fig. 6. Selected DNN operating points. Detection credits an alarm in the seven days before onset; FAR is calculated over node-days. Each point uses its own validation-selected threshold, so the plot is not an event-level precision–recall curve.

validation-selected threshold, so these are not a common operating curve.

## VIII. DISCUSSION AND LIMITATIONS

The model is a probability-ranking component, not an operational warning system. Reanalysis-derived high flow does not establish accuracy against gauges, inundation or impacts. One-day lead time must be measured from input publication times, and radar frames must be available at issue time.

The reported DNN sequence is informative for controlled representation choices, but not for a definitive model ranking. N0–N4 are single-seed comparisons, and the focal/confidence objective changes class weighting and confidence weighting together. Likewise, the five-seed configurations combine ensembling with temperature scaling. The recorded results therefore identify promising settings, not separated causal effects or statistically tested improvements. Confidence intervals, paired

resampling, and a documented test-set freeze are required before treating the AP differences as robust.

Radar evidence has a second comparability limitation. The scalar and gated-CNN variants use the focal/confidence family rather than a matched BCE control, and their reported batch sizes, coverage, and pretraining status differ. Their scores cannot isolate either the value of radar or the value of graph structure. Historical scores are development evidence: saved predictions, run manifests, post-filter sample counts, and per-horizon prediction arrays are absent.

## IX. FUTURE WORK

Future work should rerun neural and tree baselines on one fixed panel with common seeds, objectives and thresholds; retain prediction arrays for paired uncertainty intervals and calibration curves; and evaluate against independent gauges or inundation with issue-time inputs. A deployment-oriented evaluation should state an alarm policy in advance, replay only data available at the decision time, and report both episode detection and false-alarm burden at a shared operating point. This would distinguish calibration improvements from an actionable warning benefit.

## X. CONCLUSION

This study examines a terrain-conditioned deep neural network for next-day high-flow prediction at Sri Lankan river-basin locations. Recorded ablations show improvements after periodic numerical embeddings and static FiLM conditioning are introduced, while the BCE ensemble attains the highest recorded MMF-Net AP of 0.8355. Radar results remain exploratory and the available experiments do not isolate the value of graph structure. The findings support further controlled study of neural representation and fusion. Establishing a practical warning benefit requires independent flood observations, issue-time input replay and matched evaluation of onset detection, false alarms and probability calibration.

## DATA AND CODE AVAILABILITY

Code and catalogue files are available at <https://github.com/heshannethmina/Srilanka-Flood-Data-Set-Creation>. The complete training panel, checkpoints and predictions are absent from the reviewed checkout. Figure inputs and provenance accompany the manuscript.

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